Hackathon Primer — Small VLMs for Video Anomaly Detection

Where to start: prior work and fine-tuning tools examples.

1. Good Reads - VAD

1. Alert-CLIP: Abnormality-aware Latent-Enhanced Representation Tuning of CLIP for Video Anomaly Detection :Link
2. Similar Approach (uses fine-grained prompting): Link
3. Cerberus: Real-Time Video Anomaly Detection via Cascaded Vision-Language Models: Link
4. TAU-R1 (Traffic Anomaly Understanding):Link

1. Example - Fintuning Frameworks

Following are some good resources & examples:

Unsloth — fastest start. Free Colab notebooks for Qwen3-VL 8B, Gemma 3 4B, Qwen2.5-VL 7B, Llama 3.2 Vision 11B, Pixtral 12B.

example:

```
model = FastVisionModel.get_peft_model(
    model,
    finetune_vision_layers   = False,    # frozen encoder
    finetune_language_layers = True,
    r = 16, lora_alpha = 16,
    target_modules = "all-linear",
)
```

Tip: Use `UnslothVisionDataCollator` with `train_on_responses_only = True`. Build the dataset with a list comprehension, **not** `dataset.map()` — mapping breaks on multi-image samples. Refer the following doc: 📄 unsloth.ai/docs/basics/vision-fine-tuning

ms-swift — CLI-driven, broader model coverage, training + inference + eval + export in one tool.

example:

```
swift sft --model Qwen/Qwen3-VL-4B-Instruct \
    --dataset train.jsonl --val_dataset val.jsonl \
    --tuner_type lora --lora_rank 8 --lora_alpha 32 \
    --freeze_vit true --freeze_aligner true \
    --torch_dtype bfloat16 --learning_rate 1e-4 \
    --num_train_epochs 1 --per_device_train_batch_size 1 \
    --gradient_accumulation_steps 2 --gradient_checkpointing true \
    --max_length 4096 --output_dir output
```

`Tip: IMAGE_MAX_TOKEN_NUM` and `FPS_MAX_FRAMES` are your memory and latency dials.
📄 swift.readthedocs.io · dataset formats · Qwen3-VL guide

HF TRL + PEFT — the reference stack. Set `max_length=None` in `SFTConfig` or truncation silently cuts image tokens. 📄 TRL VLM SFT · HF Cookbook